Database Management System Lab

CIC-256

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MAHARAJA AGRASEN INSTITUTE OF TECHNOLOGY

VISION

"To attain global excellence through education, innovation, research, and work ethics with the commitment to serve humanity."

MISSION

The Institute shall endeavour to incorporate the following basic missions in the teaching methodology:

M1. To promote diversification by adopting advancement in science, technology, management, and allied discipline through continuous learning

M2. To foster moral values in students and equip them for developing sustainable solutions to serve both national and global needs in society and industry.

M3. To digitize educational resources and process for enhanced teaching and effective learning.

M4. To cultivate an environment supporting incubation, product development, technology transfer, capacity building and entrepreneurship.

M5. To encourage faculty-student networking with alumni, industry, institutions, and other stakeholders for collective engagement.

MAHARAJA AGRASEN INSTITUTE OF TECHNOLOGY

COMPUTER SCIENCE & ENGINEERING DEPARTMENT

VISION

“To attain global excellence through education, innovation, research, and work ethics in the field of Computer Science and engineering with the commitment to serve humanity.”

MISSION

M1: To lead in the advancement of computer science and engineering through internationally recognized research and education.

M2: To prepare students for full and ethical participation in a diverse society and encourage lifelong learning.

M3: To foster development of problem solving and communication skills as an integral component of the profession.

M4: To impart knowledge, skills and cultivate an environment supporting incubation, product development, technology transfer, capacity building and entrepreneurship in the field of computer science and engineering.

M5: To encourage faculty, student’s networking with alumni, industry, institutions, and other stakeholders for collective engagement.

Program Outcomes (PO)

PO 1: Engineering knowledge - Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to solve complex engineering problems.

PO 2: Problem analysis - Identify, formulate, research literature, and analyze complex engineering problems using first principles of mathematics, natural sciences, and engineering sciences.

PO 3: Design/Development of Solutions - Design solutions for complex engineering problems and design system components or processes that meet specified needs with consideration for public health, safety, and environmental aspects.

PO 4: Conduct investigations of complex problems - Use research-based knowledge and methods including experiment design, data analysis, and synthesis to provide valid conclusions.

PO 5: Modern tool usage - Create, select, and apply appropriate techniques, resources, and modern IT tools, including prediction and modelling , to complex engineering activities with an understanding of their limitations.

PO 6: The engineer and society - Apply reasoning informed by contextual knowledge to assess societal, health, safety, legal, and cultural issues and responsibilities related to professional engineering practice.

PO 7: Environment and sustainability - Understand the impact of engineering solutions in societal and environmental contexts and demonstrate knowledge of sustainable development.

PO 8: Ethics - Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.

PO 9: Individual and team work - Function effectively as an individual, a team member, or a leader in diverse and multidisciplinary settings.

PO 10: Communication - Communicate effectively on complex engineering activities with both the engineering community and society through reports, documentation, presentations, and clear instructions.

PO 11: Project management and finance - Demonstrate knowledge and understanding of engineering and management principles and apply them to one's work in project management and multidisciplinary environments.

PO 12: Life-long learning - Recognize the need for and possess the ability to engage in independent and life-long learning in the context of technological change.

Program Educational Objectives (PEO)

PEO1: Graduates will work with the top institutions and researchers, dedicating themselves to lifelong learning and social responsibility. (M1, M2)

PEO2: Graduates will exhibit outstanding communication skills and the capacity to collaborate effectively within diverse teams. (M3)

PEO3: Graduates cultivating skills in computer science and engineering contribute to driving innovation, entrepreneurship, and economic growth. (M4)

PEO4: Graduates network with stakeholders to contribute to the growth of the department. (M5)

Program Specific Outcome (PSO)

PSO1: Able to explore and apply emerging technologies in computer science and engineering such as Artificial Intelligence, Machine Learning, Data Science, etc.

PSO2: Able to independently and collaboratively design, develop and evaluate innovative solutions to existing problems, addressing the needs of industry and society.

PSO3: Able to pursue advanced studies, conduct research and development, and cultivate entrepreneurship skills in the modern computing environment.

INTRODUCTION TO THE LAB

Course Objectives:

1. To introduce basic concepts, architecture and characteristics of database systems
2. To introduce relational model concepts and PL/SQL programming
3. To introduce relational database design and Normal forms based on functional dependencies
4. To introduce concepts of object-oriented & distributed databases

Course Outcomes:

CO 1: Ability to understand the basics of database systems
CO 2: Ability to use SQL as DDL, DCL and DML
CO 3: Ability to understand and apply relational model concepts, SQL functions, advanced query operations, and transaction control for efficient database management.
CO 4: Ability to understand and apply relational database design principles, normalization, transaction management, concurrency control, recovery techniques, and PL/SQL programming for efficient and reliable database systems.

LIST OF EXPERIMENTS (As prescribed by G.G.S.I.P.U)

Paper Code(s): CIC-256	L	P	C
Paper: Database Management System Lab	-	2	1

Marking Scheme:
1. Teachers Continuous Evaluation: 40 marks
2. Term end Theory Examinations: 60 marks

Instructions:
1. The course objectives and course outcomes are identical to that of (Database Management System) as this is the practical component of the corresponding theory paper.
2. The practical list shall be notified by the teacher in the first week of the class commencement under intimation to the office of the Head of Department / Institution in which the paper is being offered from the list of practicals below. Atleast 10 experiments must be performed by the students, they may be asked to do more. Atleast 5 experiments must be from the given list.

1. Experiments based on DDL commands – CREATE, ALTER, DROP and TRUNCATE.
2. Apply the integrity constraints like Primary Key, Foreign key, Check, NOT NULL, etc. to the tables.
3. Experiments based on basic DML commands – SELECT, INSERT, UPDATE and DELETE.
4. Write the queries for implementing Built-in functions, GROUP BY, HAVING and ORDER BY.
5. Write the queries to implement the joins.
6. Write the queries to implement the subqueries.
7. Write the queries to implement the set operations.
8. Write the queries to create the views and queries based on views.
9. Demonstrate the concept of Control Structures.
10. Demonstrate the concept of Exception Handling.
11. Demonstrate the concept of Functions and Procedures.
12. Demonstrate the concept of Triggers.

LIST OF EXPERIMENTS (beyond syllabus)

1. The Institute offers multiple B. Tech Courses. Students can take admission in any Course offered by the Institute. Multiple subjects are taught in each branch. The institute has a Director and Hod of each B. Tech Course. The HoD of another branch may be given additional responsibility to take charge of one more department for a limited period in the absence of Hod of that branch. Faculties recruited in each B. Tech Course can teach only a single same subject to students of different B.Tech Courses. For each subject, there is a faculty coordinator responsible for coordinating all the faculties teaching the same subject. A batch of 20 students is assigned a faculty mentor. The mentor takes counsellor meetings twice a month.

    Data to be recorded in the database include:
    Student – enrollment no, name, data of birth, address, contact no, email – id, qualifying exam rank, Course
    Faculty – Faculty name, contact no, email id, course
    Parents data – Parents name, contact no, email id, Profession

    Draw the ER diagram (Conceptual design) of the given problem statement. (CO1)

2. To demonstrate a connection to a MariaDB database using the following languages: C, Java and Python

MARKING SCHEME FOR THE PRACTICAL EXAMS

There will be two practical exams in each semester.

i. Internal Practical Exam

ii. External Practical Exam

INTERNAL PRACTICAL EXAM

It is taken by the respective faculty of the batch.

MARKING SCHEME FOR THIS EXAM IS:

Total Marks: 40

Rubrics		10 Marks			POs and PSOs Covered	
		0 Marks	1 Marks	2 Marks	PO	PSO
R1	Is able to identify and define the objective of the given problem?	No	Partially	Completely	PO1, PO2	PSO1, PSO2
R2	Is proposed design/procedure/algorithm solves the problem?	No	Partially	Completely	PO1, PO2, PO3	PSO1, PSO2
R3	Has the understanding of the tool/programming language to implement the proposed solution?	No	Partially	Completely	PO1, PO3, PO5	PSO1, PSO2
R4	Are the result(s) verified using sufficient test data to support the conclusions?	No	Partially	Completely	PO2, PO4, PO5	PSO2
R5	Individuality of submission?	No	Partially	Completely	PO8, PO12	PSO1, PSO3

S.No	Programs	R1	R2	R3	R4	R5	Total Marks (10)	Date	Faculty Signature
		Is able to identify and define the objective of the given problem?	Is proposed design /procedure /algorithm solves the problem?	Has the understanding of the tool/programming language to implement the proposed solution?	Are the result(s) verified using sufficient test data to support the conclusions?	Individuality of submission?			
		2 Marks	2 Marks	2 Marks	2 Marks	2 Marks			

Experiment 1: Different Types of Databases and Their Differences

Aim:

To study different types of databases such as MySQL, PostgreSQL, MariaDB, Oracle, and MongoDB and compare their features.

Theory:

Databases are structured collections of data that allow efficient storage, retrieval, and management of information. Different database management systems (DBMS) are designed for different types of applications based on performance, scalability, and flexibility requirements.

Types of Databases

1. MySQL

Type: Relational Database Management System (RDBMS)

MySQL is one of the most widely used open-source relational databases. It uses Structured Query Language (SQL) for managing data and is commonly used in web development.

Features:

- Open source and easy to use
- High performance for web applications
- Supports ACID properties
- Widely supported by hosting providers

Applications:

- Web applications
- Content management systems
- E-commerce websites

2. PostgreSQL

Type: Object-Relational Database

PostgreSQL is an advanced open-source relational database known for its strong standards compliance and powerful features.

Features:

- Highly reliable and extensible
- Advanced indexing and query optimization
- Strong data integrity
- Supports JSON and complex queries

Applications:

- Data analytics
- Enterprise applications
- Geographic Information Systems (GIS)

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3. MariaDB

Type: Relational Database
Origin: Fork of MySQL

MariaDB was developed as a community-driven alternative to MySQL. It is fully compatible with MySQL but offers better performance in many cases.

Features:

- Open source and community maintained
- Drop-in replacement for MySQL
- Improved performance
- Enhanced storage engines

Applications:

- Web applications
- Cloud environments
- MySQL replacement systems

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4. Oracle Database

Type: Enterprise Relational Database

Oracle Database is a commercial DBMS widely used in large enterprises where high performance, security, and scalability are required.

Features:

- Very high scalability
- Strong security mechanisms
- Advanced backup and recovery
- Supports large databases

Applications:

- Banking systems
- Large enterprise software
- Mission-critical systems

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5. MongoDB

Type: NoSQL Document Database

MongoDB is a non-relational database that stores data in JSON-like documents. It is designed for handling large volumes of unstructured data.

Features:

- Schema-less design
- Highly scalable
- Flexible data model
- Good for big data applications

Applications:

- Real-time analytics
- Big data systems
- Applications with changing schema

Feature	MySQL	PostgreSQL	MariaDB	Oracle	MongoDB
Database Type	Relational	Object-Relational	Relational	Relational	No Sql Document
License	Open Source	Open Source	Open Source	Closed	Open Source
Schema	Fixed	Fixed	Fixed	Fixed	Flexible
Performance	Good	Very Good	Very Fast	Exellent	High
Scalability	Moderate	High	High	Very High	Very High
Best use case	Web Apps	Complex Systems	MySQL Replacement	Enterprises	Big Data
Query Langauge	SQL	SQl	SQL	SQL	MongoDB Query Language

Result

Thus, different types of databases were studied and compared successfully based on their features, performance, and use cases.

Attendance Management System using Database Management System

Objective:

The Attendance Management System is designed to:

- Digitally record and manage student attendance
- Reduce manual errors and paperwork
- Provide quick retrieval of attendance reports
- Ensure data consistency using DBMS concepts

Scope of the Project:

- Manage students, faculty, subjects, and classes
- Record daily attendance
- Generate attendance reports
- Enforce integrity constraints using DBMS features

Database Design:

Entities

- Student
- Faculty
- Subject
- Course/Class
- Attendance

Student Table

```
Database changed
mysql> CREATE TABLE Student (
    ->     student_id INT PRIMARY KEY,
    ->     name VARCHAR(50),
    ->     roll_no VARCHAR(20) UNIQUE,
    ->     course VARCHAR(50),
    ->     semester INT
    -> );
Query OK, 0 rows affected (0.13 sec)
```

Faculty Table

```
mysql> CREATE TABLE Faculty (
    ->     faculty_id INT PRIMARY KEY,
    ->     name VARCHAR(50),
    ->     department VARCHAR(50)
    -> );
Query OK, 0 rows affected (0.82 sec)
```

Subject Table

```
mysql> CREATE TABLE Subject (
    ->     subject_id INT PRIMARY KEY,
    ->     subject_name VARCHAR(50),
    ->     faculty_id INT,
    ->     FOREIGN KEY (faculty_id) REFERENCES Faculty(faculty_id)
    -> );
Query OK, 0 rows affected (3.50 sec)
```